

On Computational VSS for General Access Structures

Shahla Atapoor¹, Karim Baghery¹, Robin Jadoul^{2,1}, and Barry Van Leeuwen¹

¹ COSIC, KU Leuven, Leuven, Belgium,

² 3MI Labs, Leuven, Belgium.

{firstname.lastname}@kuleuven.be

Abstract. Verifiable Secret Sharing (VSS) schemes are fundamental building blocks in distributed cryptography. While most existing works focus on threshold structures, many real-world applications require more general access structures, where participants have different levels of power and only certain subsets are authorized to reconstruct the secret. Existing computational VSS schemes for general access structures typically rely on Discrete Logarithm (DL)-based homomorphic commitments, which limits their applicability, particularly in scenarios requiring Post-Quantum (PQ) security. In this work, we present a generalized version of Π , a unified framework introduced at PKC 2025 for constructing computational VSS schemes without relying on homomorphic commitments. Our framework supports arbitrary monotone $\mathcal{Q}_2$ access structures, encompassing replicated and threshold secret sharing (e.g., Shamir’s scheme), while preserving the efficiency and modularity of Π . Notably, it requires only a random oracle and any commitment scheme satisfying hiding and binding, making it compatible with a wide range of instantiations, including PQ-secure commitments. In particular, our hash-based instantiation yields the first symmetric-key-based VSS scheme for general access structures. Compared to prior general-access VSS schemes based on homomorphic commitments (e.g., variants of Pedersen scheme from FC 2003), our DL-based constructions eliminate the need for homomorphic commitments and achieve asymptotic improvements in verification and reconstruction costs. We believe that this extension enhances the versatility of the original Π framework and paves the way for its deployment in a broader range of practical distributed systems.

Keywords: Verifiable Secret Sharing · General Access Structure · Π Framework

1 Introduction

Secret sharing is a fundamental primitive in distributed cryptography, allowing a dealer to distribute a secret among multiple parties so that only designated coalitions of participants can reconstruct it. The most widely studied instance is

(n, t) -threshold secret sharing, introduced independently by Shamir [Sha79] and Blakley [Bla79], where any subset of at least $t+1$ participants can reconstruct the secret, while smaller subsets learn nothing. Although threshold schemes capture many scenarios, real-world access policies are often more nuanced and cannot be expressed by a single threshold parameter. This motivates the study of *general access structures*, in which participants have different levels of authority and only specific subsets are authorized to recover the secret. In this setting, the family of qualified coalitions is defined by an arbitrary monotone set system or Boolean function, giving general secret sharing the flexibility needed to model complex trust and authorization policies.

Basic secret sharing schemes guarantee *privacy* and *reconstructability* against only passive adversaries; they do not protect against malicious participants during secret reconstruction or against a misbehaving dealer distributing inconsistent shares. In adversarial environments, this omission undermines higher-level protocols. *Verifiable secret sharing* (VSS) schemes [Fel88, Ped92, ABCP23b, CCG24, Bag25] strengthen secret sharing by providing mechanisms to verify that the distributed shares are consistent with an underlying secret. This property is essential for building robust distributed primitives such as Distributed Key Generation (DKG) [GJKR07, ABCP23a], threshold signatures [GJKR07, ABCP23b], Multi-Party Computation (MPC) [BGW88, RB89], public randomness beacons [CD20], robust federated learning [SGA21], and many other advanced cryptographic protocols where the correctness of the underlying sharing must be assured despite faults or active adversaries.

In the synchronous communication model, early VSS constructions such as those by Feldman and Pedersen [Fel88, Ped92] are designed for threshold structures using Discrete-Logarithm (DL)-based homomorphic commitments. Extending such constructions to arbitrary access structures is nontrivial. The framework proposed by Herranz and Saez [HS03] was the first to lift Pedersen–Feldman style verifiable sharing to general access structures. It formalizes the security requirements for secrecy and robustness against an arbitrary adversary structure and demonstrates how to combine linear secret-sharing realizations of an access structure with group commitments and local consistency checks. This established that computationally secure VSS for arbitrary access structures is possible, though at the cost of expensive algebraic operations and reliance on DL assumptions, which limits applicability in scenarios requiring *Post-Quantum (PQ) security*.

For many years, it was an open question whether practical computationally secure VSS could be realized using only lightweight, hash-based commitments. Such an approach would reduce computational cost and enable post-quantum instantiations. This question was recently addressed by Atapoor, Bagheri, Cozzo, and Pedersen [ABCP23b], who proposed an efficient hash-based VSS protocol using Zero-Knowledge (ZK) proofs over secret-shared data [BBC⁺19], achieving optimal resilience in the synchronous model. Shoup and Smart [SS24] also proposed an alternative hash-based VSS for the asynchronous model. Casscudo,

Cozzo, and Giunta [CCG24] further refined the approach of [ABCP23b] by reducing broadcast communication in the optimistic case, at the expense of increased private communication. However, in distributed settings where parties simultaneously act as both dealer and verifier, their construction can impose higher communication overhead on the participants [CCG24]. Building on similar ideas, Bagheri [Bag25] introduced the unified Π framework, which abstracts and modularizes the construction of computationally secure VSS schemes in the synchronous communication model. The Π framework supports both classical and PQ instantiations, though in its original form it is tied to Shamir’s polynomial secret sharing and thus limited to threshold access structures.

Despite these advances, a gap remains regarding the construction of hash-based VSS schemes for *general access structures*. Bridging this gap would allow the use of lightweight (i.e., symmetric-key based) primitives to construct practical and PQ secure distributed protocols for various access structures, including Shamir, replicated, weighted, and others.

Our Contributions. In this paper, we close this gap by introducing a generalized variant of the Π framework [Bag25], called $G\Pi$, which enables the construction of computationally secure VSS protocols for *general access structures* in the synchronous communication model. The new framework $G\Pi$ operates under the standard Q_2 condition, which is equivalent to the honest-majority assumption in the threshold setting for which Π was originally designed. $G\Pi$ preserves the efficiency and modularity of Π while substantially broadening its scope and applicability. In particular, it allows the construction of computationally secure VSS protocols for any linear secret-sharing realization of a monotone access structure, including well-known cases such as Shamir and replicated secret sharing. From a technical perspective, $G\Pi$ relies only on the random oracle model and a standard commitment scheme satisfying basic hiding and binding properties. It thus avoids the specialized homomorphic commitments used in earlier constructions [HS03], making it possible to instantiate the framework with a wide range of commitments, including PQ-secure hash-based schemes. By combining the flexibility of general access structures with the efficiency of lightweight cryptography, $G\Pi$ achieves the essential properties of verifiability, privacy, and robustness without sacrificing practicality.

New VSS Protocols for General Access Structures. The unified framework $G\Pi$ can be instantiated in a manner similar to the original Π protocol. Using $G\Pi$ with three different commitment schemes, we construct three new VSS protocols ($G\Pi_F$, $G\Pi_P$, $G\Pi_{LA}$), each offering distinct security and efficiency trade-offs. These new VSS schemes can be seen as generalized versions of the threshold VSS protocols (Π_F , Π_P , Π_{LA}) introduced in the original Π framework [Bag25]. Analogous to Feldman’s VSS and the Π_F protocol [Fel88, Bag25], the scheme $G\Pi_F$ requires a high-entropy secret and achieves security based on the DL assumption. Similar to Pedersen’s VSS and the Π_P protocol [Ped92, Bag25], our proposed scheme $G\Pi_P$ achieves privacy against computationally unbounded adversaries, while its verifiability reduces to the DL problem. Compared to Her-

ranz and Saez’s VSS scheme for general access structures [HS03], GII_P achieves asymptotically better performance in both verification and reconstruction, at the expense of a constant slowdown of about $2\text{--}2.5\times$ in the sharing phase and broadcast communication. Therefore, we believe that most protocols built upon their construction can be further improved by employing our protocol. Finally, our scheme GII_{LA} can be viewed as the generalization of the Π_{LA} protocol [Bag25], and, to the best of our knowledge, represents the first hash-based VSS construction for general access structures.

Efficiency Comparisons of New VSS Schemes. Table 1 summarizes the performance metrics of the proposed VSS schemes (GII_F , GII_P , GII_{LA}) and compares them with the most relevant schemes from the literature [HS03].

Table 1: A comparison of new VSS schemes with those of Herranz and Saez [HS03]. *Com.:* Communication, *DL:* Discrete Logarithm, *IT-S:* Information Theoretically Simulatability, *RO:* Random Oracle, *Plain:* Plain Model, *PV:* Private, *BC:* Broadcast, *n:* Number of parties, *d:* threshold parameter ($d \leq n$), *m:* The length of shares ($m \geq n$), *E:* Exponentiation in group $\mathbb{G}$, *M:* Multiplication in $\mathbb{G}$ or $\mathbb{Z}_q$, *H:* Hashing, $|\mathbb{G}|$: $\mathbb{G}$ element size, $|\mathbb{Z}_q|$: $\mathbb{Z}_q$ element size, $|\mathcal{H}|$: Output size of $\mathcal{H}$, $|Q|$: Qualified set size, $|s_{P_j}|$: Share size of party P_j .

VSS & Security	Sharing	Dealer’s Com.	Verification	Reconstruct
Her-Sae [HS03] (DL, Plain)	$d E$ $md M$	PV: $m \mathbb{Z}_q $ BC: $d \mathbb{G} $	$(d+1) s_{P_j} E$ $+ d s_{P_j} M$	$d s_{P_j} Q E$ $+ d s_{P_j} Q M$
Sec. 4.3, GII_F (DL, RO)	$2m E$ $2md + d M$	PV: $m \mathbb{Z}_q $ BC: $n \mathbb{G} + d \mathbb{Z}_q $	$2 s_{P_j} E +$ $md + s_{P_j} M$	$2 s_Q E +$ $2 s_Q M$
Her-Sae [HS03] (DL, Plain, IT-S)	$2d E$ $2md + d M$	PV: $2m \mathbb{Z}_q $ BC: $d \mathbb{G} $	$(d+2) s_{P_j} E$ $+ d s_{P_j} M$	$d s_{P_j} Q E$ $+ d s_{P_j} Q M$
Sec. 4.1, GII_P (DL, RO, IT-S)	$2m + n E$ $2md + d M$	PV: $(n+m) \mathbb{Z}_q $ BC: $n \mathbb{G} + d \mathbb{Z}_q $	$2 s_{P_j} E +$ $md + s_{P_j} M$	$2 s_Q E +$ $2 s_Q M$
Sec. 4.2, GII_{LA} (Hash, RO)	$n \mathcal{H}$ $2md + d s_{P_j} M$	PV: $(n+m) \mathbb{Z}_q $ BC: $n \mathcal{H} + d \mathbb{Z}_q $	$2 \mathcal{H} +$ $+ md + s_{P_j} M$	$ Q \mathcal{H} +$ $+ md s_Q M$

Outline. In Section 2, we provide an overview of some preliminary concepts. In Section 3, we present our definitions and the general construction to build computational VSS for the general access structure. Leveraging the proposed general scheme, in Section 4, we present various computationally secure VSS schemes for general access structure. Finally, in Section 5, we conclude the paper.

2 Preliminaries

Notation. We let λ denote a security parameter. A function is called *negligible* in X , written $\text{negl}(X)$, if for any constant c , there exists some X_0 , such that $f(X) < X^{-c}$ for $X > X_0$. A function that is negligible in the security parameter λ is simply called negligible. We use the assignment operator $\leftarrow$ to denote uniform sampling from a set Ξ , e.g. $x \leftarrow \Xi$. For $n \in \mathbb{N}$, we write $[n] = \{1, \dots, n\}$. Finally, all logarithms are in base 2.

2.1 Access Structures and Generalized MPC

In Linear Secret Sharing Schemes (LSSS) it is common to associate an LSSS with a Monotone Span Program (MSP) realizing the access structure, (Γ, Δ) . These definitions exist over fields, [SW19] or over rings, [JSvL22]. We recap on the relevant definitions here.

Access Structures Let $\mathcal{P} = \{P_1, P_2, \dots, P_n\}$ be the set of n parties. The set of parties that an adversary, $\mathcal{A}$, can corrupt is drawn from the access structure, (Γ, Δ) . The set Γ is the set of all qualified sets, while Δ is the set of all unqualified sets. Each of these two sets is said to be monotone: For any set X' , if $X \in \Gamma$ and $X \subseteq X'$ then $X' \in \Gamma$. Similarly, for any set Y' , if $Y \in \Delta$ and $Y' \subseteq Y$ then $Y' \in \Delta$. Moreover, a set can not be both qualified and unqualified, i.e. $\Gamma \cap \Delta = \emptyset$.

An access structure is said to be $\mathcal{Q}_\ell$ when no union of ℓ unqualified sets covers the entire set of parties. Of particular relevance are the $\mathcal{Q}_2$ access structures, which may be seen as a generalization of the honest majority condition in threshold access structures, as the complement of an unqualified set is then necessarily qualified.

Definition 1 ($\mathcal{Q}_2$ Access Structures). An access structure (Γ, Δ) is said to be $\mathcal{Q}_2$ when

$$\forall A, B \in \Delta : A \cup B \neq \mathcal{P}.$$

Lemma 1. For an access structure (Γ, Δ) , an unqualified set $U \in \Delta$, and qualified set $Q \in \Gamma$, the difference $Q \setminus \bar{U}$ is unqualified.

Proof.

$$Q \setminus \bar{U} = Q \cap U \subseteq U \in \Delta.$$

Monotone Span Programs For the coming section, let $\mathbb{K}$ be a finite field, $\mathbb{F}_q$, or prime power ring, $\mathbb{Z}_{p^k}$.

Definition 2. A Monotone Span Program, $\mathcal{M}$, is a quadruple $(\mathbb{K}, M, \varepsilon, \varphi)$ where $M \in M_{m \times d}(\mathbb{K})$ is a full-rank matrix for some m and $d \leq m$, $\varepsilon \in \mathbb{K}^d$ is an arbitrary non-zero vector called the target vector, and $\varphi : [m] \Rightarrow \mathcal{P}$ is a surjective map of the rows of M to the parties in $\mathcal{P}$. The size of $\mathcal{M}$ is defined to be m , the number of rows in M .

Note that it is not uncommon to consider φ as a function which maps a length m vector to a length m vector as

$$(\varphi(\mathbf{v}, P_i))_j = \begin{cases} v_j & \varphi(j) = P_i \\ 0 & \text{otherwise} \end{cases}$$

It is clear that this obtains the same result: It assigns the shares corresponding to row j of the matrix, M , to party P_i . We will use these definitions interchangeably and let context infer the case.

Given a set of parties, $\mathcal{S} \subseteq \mathcal{P}$, we can define the submatrix $M_{\mathcal{S}}$ as the matrix whose entries are nonzero on exactly the rows indexed by $\{i \in [m] : \varphi(i) \in \mathcal{S}\}$. Similarly, $\mathbf{s}_{\mathcal{S}} = \sum_{P \in \mathcal{S}} \varphi(\mathbf{s}, P)$ is the restriction of $\mathbf{s}$ to the parties in $\mathcal{S}$.

Definition 3. An MSP, $\mathcal{M}$, is said to compute an access structure, (Γ, Δ) , if for every set $X \subset 2^{\mathcal{P}}$ it holds that

$$\begin{aligned} A \in \Gamma &\Rightarrow \varepsilon \in \text{Im}(M_A^T) \\ A \notin \Gamma &\Rightarrow \varepsilon \notin \text{Im}(M_A^T) \end{aligned}$$

Linear Secret Sharing from Monotone Span Programs When you have a Monotone Span Program (MSP) it induces a Linear Secret Sharing Scheme (LSSS) using the method in Figure 1.

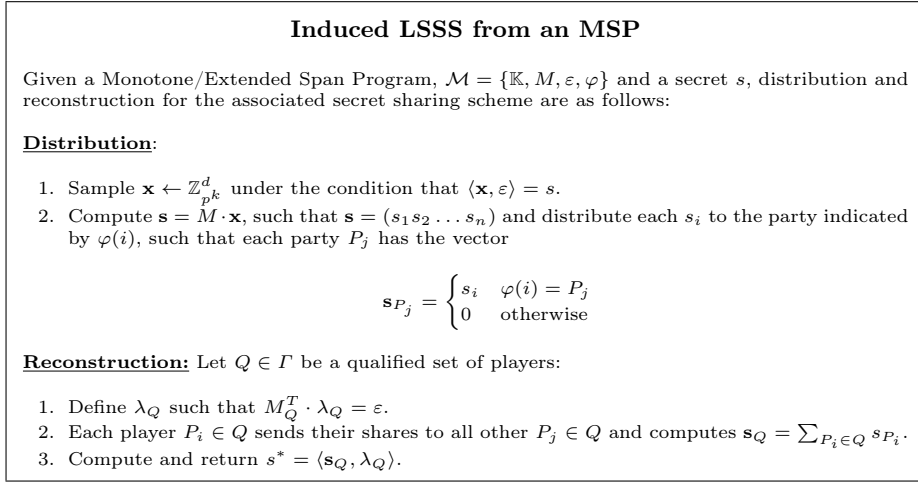


Fig. 1: Induced LSSS from a Monotone Span Program.

For any qualified set, $Q \in \Gamma$, the recombination then works as follows: As Q is qualified, according to Definition 3, there exists a vector, λ_Q , such that $M_Q^T \cdot \lambda_Q = \varepsilon$. Hence,

$$\langle \lambda_Q, \mathbf{s}_Q \rangle = \langle \lambda, \mathbf{s} \rangle = \langle \lambda, M \cdot \mathbf{x} \rangle = \langle M^T \cdot \lambda, \mathbf{x} \rangle = \langle \varepsilon, \mathbf{x} \rangle = s$$

Conversely, if $U \in \Delta$, then U is unqualified, and according to Definition 3 that means such a λ does not exist. Hence, no reconstruction can be obtained.

Example 1 (Shamir Secret Sharing expressed as an MSP). As most VSS protocols in the literature are defined explicitly for Shamir's secret sharing scheme [Fel88, Ped92, ABCP23b, CCG24, Bag25], we briefly show that this is also captured by the MSP setting. Recall that in Shamir's scheme, to share a secret s among n parties with threshold $t + 1$, one samples a random polynomial $f(x)$ of degree t whose constant term is the secret, i.e., $f(0) = s$. Then n shares are computed as $s_i = f(\alpha_i)$ for n distinct $\alpha_i \neq 0$.

Transposing this to the MSP setting, we can interpret the coefficients of f as a vector $\mathbf{f} = (f_0, f_1, \dots, f_t)$, with the inner product $\langle \mathbf{f}, (1, 0, \dots, 0) \rangle = s$, defining the target vector as $\varepsilon = (1, 0, \dots, 0)$. Polynomial evaluation in multiple points can be represented as multiplication by an $n \times (t+1)$ Vandermonde matrix M . Each row of M corresponds to a single evaluation point α_i , so φ assigns each row to the corresponding party. Summarizing this:

$$M = \begin{pmatrix} \alpha_1^0 & \alpha_1^1 & \dots & \alpha_1^t \\ \alpha_2^0 & \alpha_2^1 & \dots & \alpha_2^t \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_n^0 & \alpha_n^1 & \dots & \alpha_n^t \end{pmatrix}, \varepsilon = (1, 0, \dots, 0), \varphi(i) = P_i.$$

Example 2 (Replicated Secret Sharing). In a replicated secret sharing scheme, the secret is split into an additive sharing, where every share is assigned to all parties in the complement of a maximally unqualified set. We present an MSP for the common $(3, 1)$ -threshold case, but stress that the general construction is fully covered by the MSP formalization.

$$M = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \varepsilon = (1, 1, 1), \varphi(i) = P_{\lceil \frac{i}{2} \rceil}.$$

2.2 Vector Commitments

Vector commitments allow a committer to commit to a vector of messages in a single operation. We summarize the (basic) definition of *vector commitments*, which is a simplified version of the one presented by Catalano and Fiore [CF13].

Definition 4 (Vector Commitments). A basic vector commitment scheme over a message space $\mathcal{M}$ is a tuple of algorithms $\text{VC} = (\text{Setup}, \mathcal{C}, \text{Verify})$ defined as follows:

1. $\text{Setup}(1^\lambda) \rightarrow \text{pp}$: On input the security parameter 1^λ , this algorithm outputs public parameters pp . All subsequent algorithms implicitly take pp as input, but we omit it for notational clarity.
2. $\mathcal{C}(m_1, \dots, m_k; r) \rightarrow \text{com}$: On input a vector $(m_1, \dots, m_k) \in \mathcal{M}^k$ and a randomizer r , this algorithm outputs a commitment com .
3. $\text{Verify}(\text{com}, m_1, \dots, m_k, r) \rightarrow \{0, 1\}$: On input a commitment com , a vector $(m_1, \dots, m_k)$, and a value r , this algorithm outputs 1 (accept) or 0 (reject).

A basic vector commitment scheme must satisfy the following properties:

- **Correctness.** For security parameter λ and any vector $(m_1, \dots, m_k) \in \mathcal{M}^k$, a vector commitment scheme satisfies *perfect correctness* if:

$$\Pr \left[\text{pp} \leftarrow \text{Setup}(1^\lambda), r \leftarrow \{0, 1\}^\lambda, \text{com} \leftarrow \mathcal{C}(m_1, \dots, m_k; r) : \text{Verify}(\text{com}, m_1, \dots, m_k, r) = 1 \right] = 1.$$

- **Binding.** For security parameter λ , a basic vector commitment scheme satisfies *perfect* binding if for any unbounded adversary $\mathcal{A}$:

$$\Pr \left[\begin{array}{l} \text{pp} \leftarrow \text{Setup}(1^\lambda), (\text{com}, (\mathbf{m}, r), (\mathbf{m}', r')) \leftarrow \mathcal{A}(\text{pp}) : \\ \mathbf{m}, \mathbf{m}' \in \mathcal{M} \wedge ((\mathbf{m} \neq \mathbf{m}') \vee (r \neq r')) \wedge \\ \text{Verify}(\text{com}, \mathbf{m}, r) = 1 \wedge \text{Verify}(\text{com}, \mathbf{m}', r') = 1 \end{array} \right] = \text{negl}(\lambda).$$

For *computational* binding, this must hold against PPT adversaries.

- **Hiding.** For security parameter λ , a basic vector commitment scheme satisfies *perfect* hiding if for any unbounded adversary $\mathcal{A}$:

$$\left| \Pr \left[\begin{array}{l} \text{pp} \leftarrow \text{Setup}(1^\lambda), (\mathbf{m}_0, \mathbf{m}_1) \leftarrow \mathcal{A}(\text{pp}), b \leftarrow_{\$} \{0, 1\}, \\ r \leftarrow_{\$} \{0, 1\}^\lambda, \text{com} \leftarrow \mathcal{C}(\mathbf{m}_b; r), b' \leftarrow \mathcal{A}(\text{com}) : b = b' \end{array} \right] - \frac{1}{2} \right| = \text{negl}(\lambda).$$

For *computational* hiding, this must hold against PPT adversaries.

3 VSS for General Access Structure

In this section, we introduce a new variant of the Π framework, named $\text{G}\Pi$, that enables the construction of computationally secure VSS protocols for general access structures. We begin by presenting our formal definitions of VSS for general access structures, and then proceed to describe our proposed general construction together with its security proof.

3.1 Definitions Generalized Computational VSS

The definitions presented in [ABCP23b, Bag25] and adopted in the original Π protocol are tailored to (n, t) -threshold access structures. To develop our generalized version of Π , we extend these definitions to accommodate arbitrary general access structures.

Definition 5. A VSS generated by the MSP, $\mathcal{M} = (\mathbb{K}, M, \varepsilon, \varphi)$, realizing the (Γ, Δ) access structure consists of four PPT algorithms: Init, Share, Verify, Reconstruct.

1. $\text{Init}(\lambda) \rightarrow \text{pp}$: Takes as input the security parameter, λ , and samples the set of public parameters, $\text{pp} = (n, (\Gamma, \Delta), \mathcal{M})$. All further algorithms take pp as input in addition to the explicitly stated inputs. Then it returns the public parameters to all parties.
2. $\text{Share}(s)$: Takes as input the access structure and a secret, s , and generates the vector of shares, $\{\mathbf{s}_{P_j}\}_{j=1}^n$. Then it computes a (non-interactive) distributed proof π_{Share} , to prove the validity of the shares. Note that π_{Share} needs to be verified by a qualified set, $Q \in \Gamma$, in possession of the shares assigned to them by the underlying MSP.
3. $\text{Verify}(\mathcal{P}, \{\mathbf{s}_{P_j}\}_{j=1}^n, \pi_{\text{Share}})$: Given the set of all parties, $\mathcal{P}$, the set of shares of n parties, $\{\mathbf{s}_{P_j}\}_{j=1}^n$, and the proof, π_{Share} , the algorithm outputs **true** (and the honest parties agree on a set of correct shares) or **false**.

4. **Reconstruct**($\{\mathbf{s}_{P_i}\}_{i \in Q}, \pi_{\text{Share}}$): Takes as input a set of shares owned by a qualified set, $Q \in \Gamma$, and the proof, π_{Share} , and returns either the secret s , or false. If $Q = \mathcal{P}$ then robust reconstruction is possible.

A secure VSS scheme has the following two requirements, which are adapted from [ABCP23b, Bag25] to include the non-threshold case.

- **Verifiability constraint:** A shareholder must be able to verify the validity of the received share. If the verification return **true**, then there exists a *unique* secret s for all honest and qualified sets $Q \in \Gamma$ such that **Reconstruct** produces s . More formally, for any access structure (Γ, Δ) and for any honest and qualified set $Q \in \Gamma$, a VSS is called verifiable if for any PPT adversaries $\mathcal{A}$, and for all λ , we have:

$$\Pr \left[\text{Init}(\lambda) = pp, (\{\mathbf{s}_{P_i}\}_{i=1}^n, \pi) \leftarrow \mathcal{A}(\Gamma, \Delta), \text{Verify}(\mathcal{P}, \{\mathbf{s}_{P_i}\}_{i=1}^n, \pi) = \text{true} : \right. \\ \left. \exists s, \forall Q \in \Gamma, \text{Reconstruct}(\{\mathbf{s}_{P_j}\}_{j \in Q}, \pi) = s \right] \leq 1 - \text{negl}(\lambda)$$

- **Unpredictability:** The protocol must be unpredictable, meaning that there is no strategy in which an unqualified set $U \in \Delta$ could be compromised such that $\{\mathbf{s}_{P_j}\}_{j \in U}$ would enable someone to predict the secret s with significant advantage. More formally, for any access structure (Γ, Δ) a VSS is called unpredictable if for all PPT adversaries $\mathcal{A}$ who non-adaptively corrupts a single unqualified set $U \in \Delta$, for all λ , and a random secret s , we have:

$$\Pr \left[\text{Init}(\lambda) = pp, (\{\mathbf{s}_{P_i}\}_{i=1}^n, \pi) \leftarrow \text{Share}((\Gamma, \Delta), s); U \in \Delta; \right. \\ \left. s' \leftarrow \mathcal{A}((\Gamma, \Delta), \{\mathbf{s}_{P_j}\}_{j \in U}, \pi) : s' = s \right] \leq \text{negl}(\lambda)$$

- **Simulatability:** In some cases we can amplify the unpredictability definition by simulating the view of the corrupted parties. Given a protocol, Π , for any adversary corrupting an unqualified set, $U \in \Delta$, if the joint view of those corrupted parties can be simulated without knowledge of the secret we call the protocol, Π , *simulatable*. More formally, for any access structure (Γ, Δ) , we say a VSS protocol, Π , satisfies simulatability if for all PPT adversaries $\mathcal{A}$ corrupting a set $U \in \Delta$, for any secret $s \in \mathbb{K}$ and security parameter λ , there exists a simulator $\mathcal{S}$ such that:

$$\text{Let } (\{\mathbf{s}_{P_i}\}_{i=1}^n, \pi) \leftarrow \text{Share}((\Gamma, \Delta), s), \\ \text{Then } \mathcal{S}(U, \{\mathbf{s}_{P_j}\}_{j \in U}) \approx \text{View}_{\Pi, \mathcal{A}}(\{\mathbf{s}_{P_i}\}_{i=1}^n, \pi)$$

That is, the adversary's view—consisting of the shares of corrupted parties and any public commitment or proof—can be simulated based only on the corrupted parties' shares, without access to the secret s or the shares of honest parties. This ensures that an adversary corrupting at most t parties learns nothing beyond the shares it receives.

3.2 Construction of GII Protocol

Let D be a dealer and let $P_1, \dots, P_n = \mathcal{P}$ be the n participants. In some distributed protocols, D could be included in the protocol such that $D = P_j$ for

some $j \in [n]$. Let $\mathcal{C}$ be a secure (vector) commitment scheme as described in Section 2.2 that satisfies both binding and hiding properties which could either be perfect or computational. Moreover, let $\mathcal{M} = (\mathbb{K}, M, \varepsilon, \varphi)$ be a monotone span program. Then, our proposed construction for the generalized protocol GII is described in Figure 2.

Theorem 1 (Security of GII VSS Protocol). *If the (vector) commitment scheme $\mathcal{C}$ is perfectly (resp. computationally) hiding and computationally (resp. perfectly) binding, and $\mathcal{RO}$ is a random oracle, then the generic construction given in Figure 2 constitutes a secure VSS scheme for the $\mathcal{Q}_2$ access structures. Specifically: 1) the **Reconstruct** protocol running by any qualified set of (honest) shareholders, always yields a unique secret, 2) any non-qualified set of computationally unbounded (resp. bounded) shareholders cannot gain any information about the secret.*

Proof. As outlined in the **Verify** algorithm, the disqualification decision relies exclusively on publicly broadcast information, ensuring that all honest shareholders reach the same conclusion. Furthermore, if the dealer is honest and follows the **Share** algorithm correctly, then the **Verify** algorithm returns **true**, and each honest shareholder receives a valid and distinct share of a uniquely defined secret.

Verifiability. Assume that the adversary, $\mathcal{A}$, has chosen an unqualified set $U \in \Delta$ to corrupt. Then by the $\mathcal{Q}_2$ assumption, the compliment, $\overline{U}$, must be a qualified set in the access structure. Additionally, each party, P_j , in $\overline{U}$ is honest as the adversary can only corrupt one unqualified set. For simplicity in the notation, set $\overline{U} = Q$. Hence it is possible for at least one qualified set to reconstruct the secret - hence s is well-defined after the sharing phase.

First, we show that a party, without communication, can verify that its share is consistent with the commitment to the share of the dealer.

Let $\mathbf{x} \in \mathbb{K}^d$ and $\mathbf{x}' \in \mathbb{K}^d$ be vectors over the field $\mathbb{K}$, and define $\mathbf{z} := \mathbf{x}' + \alpha \cdot \mathbf{x} \in \mathbb{K}^d$, where the scalar α is derived from the random oracle, i.e., $\alpha = \mathcal{RO}(\mathbf{c}_{P_1}, \dots, \mathbf{c}_{P_n})$. As the $P_j \in Q$ are honest, each P_j computes $\mathbf{z}_{P_j} = \varphi(M \cdot \mathbf{z}, P_j)$ correctly. It then verifies whether the following equation holds:

$$\mathbf{c}_{P_j} \stackrel{?}{=} \mathcal{C}(\mathbf{s}_{P_j}, \mathbf{z}_{P_j} - \alpha \cdot \mathbf{s}_{P_j}, \gamma_{P_j}).$$

If the equation holds, then the shareholder has verified that its share is consistent with the commitment to the share made by the dealer.

To show that s is unique, assume that there exists another qualified set consisting solely of honest parties, Q' . We show that if Q' calls **Reconstruct** the output, s' , is equal to the well-defined s above.

Given the vector $\mathbf{z}$, along with the openings $(\mathbf{s}_{P_j}, \gamma_{P_j})$ from all parties in the qualified set, every $P_k \in Q'$ can compute

$$(\mathbf{z}_{P_j})_i = (\varphi(M \cdot \mathbf{z}, P_j))_i = \begin{cases} z_i & \varphi(i) = P_j \\ 0 & \text{otherwise} \end{cases}$$

GII: The Generalized Variant of Π Protocol

For the protocol assume the public knowledge of the access structure (Γ, Δ) and the monotone span program $\mathcal{M} = (\mathbb{K}, M, \varepsilon, \varphi)$. Moreover, let $\mathcal{RO}$ be a random oracle. These can be considered inputs to the protocols without explicit statement.

Init(λ): Takes as input the security parameter, λ , and samples the set of public parameters, $pp = (n, (\Gamma, \Delta), \mathcal{M})$. All further algorithms take pp as input in addition to the explicitly stated inputs. Then the parties generate parameters for vector commitment scheme $\mathcal{C}$. Finally, it returns the public parameters for $\mathcal{C}$ and pp to all parties.

Share(pp, s): Upon input of the secret s , the dealer D does the following:

1. Sample two random vectors $\mathbf{x} \in \mathbb{K}^d$ and $\mathbf{x}' \in \mathbb{K}^d$, such that $\langle \mathbf{x}, \varepsilon \rangle = s$, where ε is the target vector of the MSP, and $\langle \mathbf{x}', \varepsilon \rangle = r$ where r is some random element of $\mathbb{K}$.
2. Compute $\mathbf{s} = M \cdot \mathbf{x}$ and $\mathbf{r} = M \cdot \mathbf{x}'$, such that $\mathbf{s} = (s_1, \dots, s_m)$ and $\mathbf{r} = (r_1, \dots, r_m)$. Then, assign each s_i and r_i to the parties indicated by $\varphi(i)$ such that each party P_j , for $j \in [n]$ is assigned the vectors

$$(\mathbf{s}_{P_j})_i = \begin{cases} s_i & \varphi(i) = P_j \\ 0 & \text{otherwise} \end{cases} \quad \text{and} \quad (\mathbf{r}_{P_j})_i = \begin{cases} r_i & \varphi(i) = P_j \\ 0 & \text{otherwise} \end{cases}.$$

3. For $j \in [n]$, compute $\mathbf{c}_{P_j} = \mathcal{C}(\mathbf{s}_{P_j}, \mathbf{r}_{P_j}, \gamma_{P_j})$, where $\gamma_{P_j} \in \mathbb{K}$ are the additional randomnesses used in the commitment phase, and $(\mathbf{s}_{P_j}, \mathbf{r}_{P_j})$ are the shares associated with party P_j .
4. Compute $\alpha = \mathcal{RO}(\mathbf{c}_{P_1}, \dots, \mathbf{c}_{P_n})$ and the vector $\mathbf{z} = \mathbf{x}' + \alpha \cdot \mathbf{x}$.
5. Then, set $\pi_{\text{Share}} = (\mathbf{c}_{P_1}, \dots, \mathbf{c}_{P_n}, \mathbf{z})$.
6. Send $(\mathbf{s}_{P_j}, \gamma_{P_j})$ privately to each party P_j for $j \in [n]$ and broadcast π_{Share} .

Verify($pp, \pi_{\text{Share}}, \{\mathbf{s}_{P_j}\}_{j=1}^n$): Given π_{Share} and $\{\mathbf{s}_{P_j}\}_{j=1}^n$, parties act as follows:

1. For $j \in [n]$, each party P_j first computes:

$$(\mathbf{z}_{P_j})_i = \varphi(M \cdot \mathbf{z}, P_j) = \begin{cases} z_i & \varphi(i) = P_j \\ 0 & \text{otherwise} \end{cases}.$$

2. Then, P_j computes $\alpha = \mathcal{RO}(\mathbf{c}_{P_1}, \dots, \mathbf{c}_{P_n})$, and checks if

$$\mathbf{c}_{P_j} \stackrel{?}{=} \mathcal{C}(\mathbf{s}_{P_j}, \mathbf{z}_{P_j} - \alpha \cdot \mathbf{s}_{P_j}, \gamma_{P_j}).$$

3. If the verification of P_j fails, then P_j broadcasts a complaint against the dealer. If the set of complaining parties forms a qualified set, the dealer is disqualified and the verification process outputs false.
4. In case P_j raises a complaint about the verification of their part, the dealer will broadcast $(\mathbf{s}_{P_j}, \gamma_{P_j})$ to enable everyone to verify it using the verification equation. If the verification passes, the protocol continues as usual. However, if it fails, the dealer will be disqualified, leading to a false verification outcome. Since the disqualification decision is solely based on the information broadcasted, all honest parties will ultimately reach a consensus either on a set of qualified parties Q or on rejecting the final verification.

Reconstruct($pp, \pi_{\text{Share}}, Q$): Given the proof π_{Share} and a qualified set, $Q \in \Gamma$, do the following:

1. Let λ_Q be the recombination vector assigned to Q such that $M_Q^T \cdot \lambda_Q = \varepsilon$.
2. Each party $P_j \in Q$ broadcasts the valid shares along with the corresponding randomnesses of $(\mathbf{s}_{P_j}, \gamma_{P_j})_{j \in Q}$.
3. Each party computes $\alpha = \mathcal{RO}(\mathbf{c}_{P_1}, \dots, \mathbf{c}_{P_n})$.
4. Then for all shares received from other parties checks if $\mathbf{c}_{P_j} \stackrel{?}{=} \mathcal{C}(\mathbf{s}_{P_j}, \mathbf{z}_{P_j} - \alpha \cdot \mathbf{s}_{P_j}, \gamma_{P_j})$, where $j \in Q$. If all the checks passes then the protocol continue otherwise abort.
5. Then using all shares they compute $\mathbf{s}_Q = \sum_{P_i \in Q} \mathbf{s}_{P_i}$.
6. Compute and return $s^* = \langle \mathbf{s}_Q, \lambda_Q \rangle$.

Fig. 2: GII: A unified framework for building VSS schemes for general access structure.

for all $P_j \in Q'$, and then compute $\alpha = \mathcal{RO}(\mathbf{c}_{P_j})$ and after that check if the equation $\mathbf{c}_{P_j} \stackrel{?}{=} \text{Commit}(\mathbf{s}_{P_j}, \mathbf{z}_{P_j} - \alpha \cdot \mathbf{s}_{P_j}, \gamma_{P_j})$ holds for all the parties in the qualified set. Then using all shares it is possible to compute the vector of $\mathbf{s}_{Q'} = \sum_{P_j \in Q'} s_{P_j}$ along with $s_i = \langle \mathbf{s}_{Q'}, \lambda_{Q'} \rangle$, where $\lambda_{Q'}$ is the recombination vector assigned to Q' .

For contradiction, suppose that $s' \neq s$. Then, as M is full rank,

$$s' \neq s \Rightarrow \langle \mathbf{x}', \varepsilon \rangle \neq \langle \mathbf{x}, \varepsilon \rangle \Rightarrow M\mathbf{x}' \neq M\mathbf{x} \Rightarrow \mathbf{s}' \neq \mathbf{s}$$

holds. Now consider $Q^* = Q' \setminus Q$. By our choice of Q and Lemma 1, Q^* is unqualified, hence from [SW19, Lemma 1], we know that $\varphi(\text{supp}(\mathbf{s}'_{Q \cup Q'} - \mathbf{s}_{Q \cup Q'})) \supset Q^*$. Therefore, there exists a difference $\mathbf{s}_{P_i}$ and $\mathbf{s}'_{P_i}$ for some $P_i \in Q \cap Q'$. This implies that

$$\mathcal{C}(\mathbf{s}_{P_i}, \mathbf{r}_{P_i}, \gamma_{P_i}) = \mathcal{C}(\mathbf{s}'_{P_i}, \mathbf{r}'_{P_i}, \gamma_{P_i})$$

which is a contradiction with the binding property of the commitment. Hence, Q and Q' must have reconstructed a unique identical s .

Simulatability. Since the dealer is honest, the shares $\mathbf{s}_U$ are consistent with a secret sharing $\mathbf{s}$. By [SW19, Lemma 2] or its generalization from [JSvL22], we know that a matrix N exists such that $N \cdot \mathbf{s} = 0$, and so $N \cdot (\mathbf{s}_U + \mathbf{s}_{P \setminus U}) = 0$. At least one solution $\mathbf{s}_{P \setminus U}^*$ exists, and without loss of generality, we can write all $\mathbf{s}_{P \setminus U} = \mathbf{s}_{P \setminus U}^* + \mathbf{v}$ for some vector $\mathbf{v}$ in the kernel of $N_{P \setminus U}$. The simulator now samples this $\mathbf{v}$ uniformly at random, subject to the constraint that $\langle \mathbf{s}, \lambda_P \rangle = s$.

From here on out, the simulator can execute logic of the dealer and the honest parties normally, with the above $\mathbf{s}$ as share vector. Since the commitment scheme is perfectly (resp. computationally) hiding, the distributions of the commitments $\mathcal{C}(\cdot, \gamma)$ are perfectly (resp. computationally) indistinguishable, and the resulting transcript is perfectly (resp. computationally) indistinguishable, and simulatability holds. $\square$

Remark 1. The Verifiability Proof can, like in the original Π [Bag25], be seen as the special soundness of a Σ -protocol proof of knowledge with designated verification for the randomness vector $\mathbf{x}$.

4 Practical VSS Schemes for General Access Structure

The flexibility of the GII protocol (depicted in Figure 2) stems from its simplicity and broad applicability. It requires only a random oracle $\mathcal{H}$ and a secure vector commitment scheme $\mathcal{C}$, without imposing the requirement that the commitment scheme be homomorphic. Consequently, this framework can be instantiated in a manner analogous to the original Π construction, while supporting VSS schemes for general access structures. In practice, random oracles are typically instantiated by secure cryptographic hash functions, while (basic) vector commitments can be realized using a variety of cryptographic techniques grounded in different hardness assumptions.

In this section, we present concrete instantiations of GII using both DL-based vector commitment schemes and hash-function-based constructions, leading to novel VSS protocols for general access structures. In each case, the random oracle $\mathcal{RO}$ is instantiated with a suitably defined hash function $\mathcal{H}$. The resulting protocols can be viewed as extensions of the VSS constructions built by the original Π protocol [Bag25], now generalized to support general access structures.

4.1 GII_P : IT-Secure VSS Scheme from Pedersen Commitment

In our initial instantiation of GII, we employ a variant of the Pedersen vector commitment using multiple random group generators $(\mathbf{g}_1, \mathbf{g}_2, g_3)$ in a group $\mathbb{G}$ of prime order q , i.e.,

$$\mathcal{C}(\mathbf{m}, \mathbf{r}, \gamma) = \mathbf{g}_1^{\mathbf{m}} \mathbf{g}_2^{\mathbf{r}} g_3^{\gamma}.$$

As in the original Pedersen commitment, this construction achieves *perfect hiding* and *computational binding* under the discrete logarithm assumption. This choice gives rise to a new VSS scheme that ensures perfect simulatability —providing *information-theoretic privacy* of the secret against computationally unbounded adversaries— while *verifiability* remains based on the discrete logarithm assumption in $\mathbb{G}$, offering *computational security* against classical adversaries. This VSS scheme can be used in the construction of fully secure DKG protocols or distributed randomness generation schemes, where the generated values must be uniformly random against both unbounded and computationally bounded adversaries [GJKR07, CD20]. Next, we describe the steps of the new VSS scheme GII_P , which instantiates the general GII framework.

Share $(pp, \mathbf{g}_1, \mathbf{g}_2, g_3, s)$: Given the set of random group generators $(\mathbf{g}_1, \mathbf{g}_2, g_3)$ —where $(\mathbf{g}_1, \mathbf{g}_2)$ may themselves be vectors of random generators— and the public parameters pp , the dealer shares a secret s by following the steps in Figure 2, with the following modifications:

3. For $j \in [n]$ compute $\mathbf{c}_{P_j} = \mathbf{g}_1^{s_{P_j}} \mathbf{g}_2^{\mathbf{r}_{P_j}} g_3^{\gamma_{P_j}}$, where $\gamma_{P_j} \in \mathbb{K}$ are the additional randomnesses used in the commitment phase, and $(s_{P_j}, \mathbf{r}_{P_j})$ are the shares associated with party P_j .

Verify $(pp, \mathbf{g}_1, \mathbf{g}_2, g_3, \pi_{\text{Share}}, \{s_{P_j}\}_{j=1}^n)$:

2. Then, P_j computes $\alpha = \mathcal{RO}(\mathbf{c}_{P_1}, \dots, \mathbf{c}_{P_n})$, and checks if

$$\mathbf{c}_{P_j} \stackrel{?}{=} \mathbf{g}_1^{s_{P_j}} \mathbf{g}_2^{(\mathbf{z}_{P_j} - \alpha \cdot \mathbf{s}_{P_j})} g_3^{\gamma_{P_j}}.$$

Reconstruct $(pp, \mathbf{g}_1, \mathbf{g}_2, g_3, \pi_{\text{Share}}, Q)$:

- 4: Then for all shares received from other parties checks if $\mathbf{c}_{P_j} \stackrel{?}{=} \mathbf{g}_1^{s_{P_j}} \mathbf{g}_2^{(\mathbf{z}_{P_j} - \alpha \cdot \mathbf{s}_{P_j})} g_3^{\gamma_{P_j}}$, where $j \in Q$. If all the checks passes then the protocol continue otherwise abort.

4.2 GII_{LA} : VSS Scheme from Hash Functions

In contrast to our first instantiation, where the resulting VSS scheme GII_{P} relied on a Pedersen-type commitment whose verifiability is based on the DL assumption—and thus insecure against quantum adversaries—we now instantiate GII with a hash-based commitment scheme, i.e.,

$$\mathcal{C}(\mathbf{m}, \mathbf{r}, \gamma) = \mathbf{H}(\mathbf{m}, \mathbf{r}, \gamma),$$

where $\mathbf{H}$ is a secure cryptographic hash function. This leads to the first symmetric-key-based VSS scheme for general access structures, which we denote by GII_{LA} . Similar to the Shamir-based VSS scheme Π_{LA} [Bag25], our construction requires only lightweight cryptographic operations. Since modern hash functions are widely believed to remain secure even against quantum adversaries, GII_{LA} plausibly achieves *post-quantum security*. Beyond its conceptual novelty, the scheme can improve the efficiency of large-scale protocols and serve as a building block for PQ-secure distributed protocols. Below, we describe the steps of the new VSS scheme GII_{P} , which instantiates the general GII framework.

Share($pp, \mathbf{H}, \mathcal{H}, s$): Given a secure hash function $\mathbf{H}$, and the public parameters pp , to share a secret s , the dealer follows the steps outlined in Figure 2, specifically, with the following deviations:

3. For $j \in [n]$ compute $\mathbf{c}_{P_j} = \mathbf{H}(\mathbf{s}_{P_j}, \mathbf{r}_{P_j}, \gamma_{P_j})$, where $\gamma_{P_j} \in \mathbb{K}$ are the randomnesses used in the commitment phase, and $(\mathbf{s}_{P_j}, \mathbf{r}_{P_j})$ are the shares associated with party P_j .

Verify($pp, \mathbf{H}, \mathcal{H}, \pi_{\text{Share}}, \{\mathbf{s}_{P_j}\}_{j=1}^n$):

2. Then, P_j computes $\alpha = \mathcal{RO}(\mathbf{c}_{P_1}, \dots, \mathbf{c}_{P_n})$, and checks if

$$\mathbf{c}_{P_j} \stackrel{?}{=} \mathbf{H}(\mathbf{s}_{P_j}, \mathbf{z}_{P_j} - \alpha \cdot \mathbf{s}_{P_j}, \gamma_{P_j}).$$

Reconstruct($pp, \mathbf{H}, \mathcal{H}, \pi_{\text{Share}}, Q$):

- 4: Then for all shares received from other parties checks if $\mathbf{c}_{P_j} \stackrel{?}{=} \mathbf{H}(\mathbf{s}_{P_j}, \mathbf{z}_{P_j} - \alpha \cdot \mathbf{s}_{P_j}, \gamma_{P_j})$, where $j \in Q$. If all the checks passes then the protocol continue otherwise abort.

4.3 GII_{F} : DL-Based VSS Scheme for High-Entropy Secrets

Both VSS schemes presented in Sections 4.1 and 4.2 are direct instantiations of GII . We now introduce a variant of the GII_{P} VSS scheme, denoted by GII_{F} , which achieves a weaker notion of security but remains adequate in many practical settings. In particular, it can suffice for cryptographic protocols where the dealer distributes a high-entropy secret and *computational privacy* is considered sufficient, such as in distributed key generation. Compared to the GII_{P} scheme, the additional randomizer γ_{P_j} is omitted from the commitments, i.e.,

$$\mathbf{c}_{P_j} := \mathbf{g}_1^{\mathbf{s}_{P_j}} \mathbf{g}_2^{\mathbf{r}_{P_j}}.$$

The construction of GII_{F} can be derived from the description of GII_{P} , summarized in Section 4.1, by simply removing the **highlighted** terms. The following theorem shows that under the DL assumption, the VSS scheme GII_{F} satisfies *Verifiability* and *Unpredictability*, as defined in Section 3.1.

Theorem 2 (Security of $\text{G}\Pi_{\text{F}}$ VSS Protocol). Assume $(\mathbf{g}_1, \mathbf{g}_2)$ are two vectors of independent group generators for $\mathbb{G}$, and $\mathcal{H}$ behaves as a random oracle. Under the hardness of the DL problem, the VSS scheme $\text{G}\Pi_{\text{F}}$ (summarized in Section 4.1 without highlighted terms) constitutes a secure VSS scheme for the $\mathcal{Q}_2$ access structures. In particular: 1) The Reconstruct procedure always outputs the unique secret distributed by the dealer whenever a qualified honest subset of participants is involved. 2) Any unqualified set of computationally bounded shareholders cannot recover the secret.

Proof. The argument for verifiability parallels the strategy in Theorem 1.

Unpredictability. Suppose that given π_{share} and the shares of unqualified set of malicious parties, i.e., $\{\mathbf{s}_{P_j}\}_{P_j \in U}$, an adversary $\mathcal{A}$ is able to derive the secret s . We describe how to construct another adversary $\mathcal{B}$ that, by running $\mathcal{A}$ as a subroutine, can solve the DL problem.

Let (Γ, Δ) be the adversary structure for player set $\mathcal{P} = (P_1, \dots, P_n)$. The reduction proceeds as follows:

1. $\mathcal{B}$ receives a DL challenge, $(g_1, y_1 = g_1^\sigma)$. It's objective is to recover σ .
2. The adversary samples random β_i , with $\beta_1 = 1$, and sets $g_i = g_1^{\beta_i}$. For conciseness, we write $(\mathbf{g}_1, \mathbf{g}_2) = (g_1, g_2, \dots, g_{2m})$.
3. Without loss of generality we can assume that there exists a set $Q \in \Gamma$ for which it holds³ that $Q \setminus P_1 \in \Delta$. We call that set U . Next, the adversary samples uniformly at random all but one elements $\mathbf{s}_{Q,i}$ of $\mathbf{s}$ corresponding to non-zero elements in λ_Q and the first element will be the (unknown) value σ . For each of these values in $\mathbf{s}_Q$, the adversary sets $y_i = g_1^{\mathbf{s}_{Q,i}}$.
4. By a strategy similar to the approach for simulatability in Theorem 1, it is possible to compute a share vector $\mathbf{s}$ from $\mathbf{s}_Q$ using only linear operations, and this can hence be computed in the exponent of g_1 .
5. $\mathcal{B}$ selects a random vector $\mathbf{z}$ and a random challenge α and using the public matrix M , it computes

$$(\mathbf{z}_{P_i})_i = \varphi(M \cdot \mathbf{z}) = \begin{cases} z_i & \varphi(i) = P_i \\ 0 & \text{otherwise} \end{cases}$$

for each $P_i \in Q$. Then using the obtained $\mathbf{z}_{P_i}$, the adversary computes

$$\mathbf{c}_{P_i} = \prod_{\varphi(i)=P_i} \frac{y_i^{\beta_i} \cdot g_{m+i}^{\mathbf{z}_i}}{y_i^{\beta_{m+i} \cdot \alpha}}$$

6. The oracle, $\mathcal{H}$, is programmed so that $\alpha = \mathcal{H}((\mathbf{c}_{P_i})_{P_i \in Q})$.
7. $\mathcal{B}$ now sends $(\mathbf{g}_1, \mathbf{g}_2, (\mathbf{c}_{P_i})_{P_i \in \mathcal{P}}, \mathbf{z}, \mathbf{s}_U)$ to $\mathcal{A}$.
8. As this provides $\mathcal{A}$ with the shares of all malicious parties of an unqualified set, $\mathcal{A}$ returns the secret s .

³ Should $Q \setminus P_1 = \emptyset$ then consider $\mathbf{s}_{P_1}$ the share of a different party P_i and execute the same argument

9. Finally $\mathcal{B}$ retrieves λ_Q and solves for $\mathbf{s}_{P_1}$ using the linear equation

$$\langle \lambda_Q, \mathbf{s}_Q \rangle = s$$

where σ is the only unknown and sends σ as the solution to the DL challenge.

□

Efficiency . We compare the efficiency of new VSS schemes with the most relevant existing schemes, as summarized in Table 1.

5 Conclusion

We proposed $\text{G}\Pi$, a generalized variant of the Π framework [Bag25], and demonstrated its power to build computational VSS schemes for general access structures under the Q_2 condition. $\text{G}\Pi$ preserves the simplicity, efficiency, and modularity of Π , while significantly broadening its applicability beyond threshold settings. By avoiding specialized homomorphic commitments and relying only on a random oracle and a standard commitment scheme, $\text{G}\Pi$ can be instantiated with lightweight, post-quantum secure primitives. Using three different instantiations, we presented new VSS protocols ($\text{G}\Pi_F$, $\text{G}\Pi_P$, $\text{G}\Pi_{LA}$), which generalize the threshold protocols from [Bag25] and offer distinct efficiency and security trade-offs. To the best of our knowledge, $\text{G}\Pi_{LA}$ constitutes the first hash-based VSS scheme for general access structures. To illustrate the generality of our approach, we showed how well-known secret-sharing schemes such as Shamir and replicated secret sharing can be expressed as monotone span programs and thus instantiated within our framework. This demonstrates that our techniques can directly yield, for example, a hash-based verifiable replicated secret sharing protocol, or recover the Π_{LA} scheme [Bag25] when applied to the Shamir case.

Our work opens several promising directions for further exploration. Since $\text{G}\Pi$ supports instantiations with hash-based commitments, it provides a concrete pathway toward PQ-secure VSS protocols for general access structures. Exploring practical applications of such protocols, for example in MPC, randomness generation, or e-voting systems, is an important next step. While our results focus on VSS, many applications require public verifiability. An intriguing research direction is to extend our framework to obtain Publicly VSS (PVSS) [Sch99, CD20] or even Pre-Constructed PVSS [BKNR25] schemes for general access structures. Our protocols currently assume synchronous communication. Extending the techniques developed here to the asynchronous setting remains an interesting open problem.

Acknowledgments

This work has been supported in part by CyberSecurity Research Flanders with reference number VR20192203.

References

- ABCP23a. Shahla Atapoor, Karim Baghery, Daniele Cozzo, and Robi Pedersen. Practical robust DKG protocols for CSIDH. In Mehdi Tibouchi and Xiaofeng Wang, editors, *ACNS 2023: 21st International Conference on Applied Cryptography and Network Security, Part II*, volume 13906 of *Lecture Notes in Computer Science*, pages 219–247, Kyoto, Japan, June 19–22, 2023. Springer, Cham, Switzerland.
- ABCP23b. Shahla Atapoor, Karim Baghery, Daniele Cozzo, and Robi Pedersen. VSS from distributed ZK proofs and applications. In Jian Guo and Ron Steinfeld, editors, *Advances in Cryptology – ASIACRYPT 2023, Part I*, volume 14438 of *Lecture Notes in Computer Science*, pages 405–440, Guangzhou, China, December 4–8, 2023. Springer, Singapore, Singapore.
- Bag25. Karim Baghery. *II*: A unified framework for computational verifiable secret sharing. In Tibor Jager and Jiaxin Pan, editors, *PKC 2025: 28th International Conference on Theory and Practice of Public Key Cryptography, Part IV*, volume 15677 of *Lecture Notes in Computer Science*, pages 110–142, Røros, Norway, May 12–15, 2025. Springer, Cham, Switzerland.
- BBC⁺19. Dan Boneh, Elette Boyle, Henry Corrigan-Gibbs, Niv Gilboa, and Yuval Ishai. Zero-knowledge proofs on secret-shared data via fully linear PCPs. In Alexandra Boldyreva and Daniele Micciancio, editors, *Advances in Cryptology – CRYPTO 2019, Part III*, volume 11694 of *Lecture Notes in Computer Science*, pages 67–97, Santa Barbara, CA, USA, August 18–22, 2019. Springer, Cham, Switzerland.
- BGW88. Michael Ben-Or, Shafi Goldwasser, and Avi Wigderson. Completeness theorems for non-cryptographic fault-tolerant distributed computation (extended abstract). In *20th Annual ACM Symposium on Theory of Computing*, pages 1–10, Chicago, IL, USA, May 2–4, 1988. ACM Press.
- BKNR25. Karim Baghery, Noah Knapen, Georgio Nicolas, and Mahdi Rahimi. Preconstructed publicly verifiable secret sharing and applications. In Marc Fischlin and Veelasha Moonsamy, editors, *ACNS 2025: 23rd International Conference on Applied Cryptography and Network Security, Part I*, volume 15825 of *Lecture Notes in Computer Science*, pages 89–119, Munich, Germany, June 23–26, 2025. Springer, Cham, Switzerland.
- Bla79. G. R. Blakley. Safeguarding cryptographic keys. *Proceedings of AFIPS 1979 National Computer Conference*, 48:313–317, 1979.
- CCG24. Ignacio Cascudo, Daniele Cozzo, and Emanuele Giunta. Verifiable secret sharing from symmetric key cryptography with improved optimistic complexity. In Kai-Min Chung and Yu Sasaki, editors, *Advances in Cryptology – ASIACRYPT 2024, Part VII*, volume 15490 of *Lecture Notes in Computer Science*, pages 100–128, Kolkata, India, December 9–13, 2024. Springer, Singapore, Singapore.
- CD20. Ignacio Cascudo and Bernardo David. ALBATROSS: Publicly Attestable Batched Randomness based On Secret Sharing. In Shiho Moriai and Huaxiong Wang, editors, *Advances in Cryptology – ASIACRYPT 2020, Part III*, volume 12493 of *Lecture Notes in Computer Science*, pages 311–341, Daejeon, South Korea, December 7–11, 2020. Springer, Cham, Switzerland.
- CF13. Dario Catalano and Dario Fiore. Vector commitments and their applications. In Kaoru Kurosawa and Goichiro Hanaoka, editors, *PKC 2013: 16th*

- International Conference on Theory and Practice of Public Key Cryptography*, volume 7778 of *Lecture Notes in Computer Science*, pages 55–72, Nara, Japan, February 26 – March 1, 2013. Springer Berlin Heidelberg, Germany.
- Fel88. Frank A. Feldman. Fast spectral tests for measuring nonrandomness and the DES. In Carl Pomerance, editor, *Advances in Cryptology – CRYPTO’87*, volume 293 of *Lecture Notes in Computer Science*, pages 243–254, Santa Barbara, CA, USA, August 16–20, 1988. Springer Berlin Heidelberg, Germany.
- GJKR07. Rosario Gennaro, Stanislaw Jarecki, Hugo Krawczyk, and Tal Rabin. Secure distributed key generation for discrete-log based cryptosystems. *Journal of Cryptology*, 20(1):51–83, January 2007.
- HS03. Javier Herranz and Germán Sáez. Verifiable secret sharing for general access structures, with application to fully distributed proxy signatures. In Rebecca Wright, editor, *FC 2003: 7th International Conference on Financial Cryptography*, volume 2742 of *Lecture Notes in Computer Science*, pages 286–302, Guadeloupe, French West Indies, January 27–30, 2003. Springer Berlin Heidelberg, Germany.
- JSvL22. Robin Jadoul, Nigel P. Smart, and Barry van Leeuwen. MPC for Q_2 access structures over rings and fields. In Riham AlTawy and Andreas Hülsing, editors, *SAC 2021: 28th Annual International Workshop on Selected Areas in Cryptography*, volume 13203 of *Lecture Notes in Computer Science*, pages 131–151, Virtual Event, September 29 – October 1, 2022. Springer, Cham, Switzerland.
- Ped92. Torben P. Pedersen. Non-interactive and information-theoretic secure verifiable secret sharing. In Joan Feigenbaum, editor, *Advances in Cryptology – CRYPTO’91*, volume 576 of *Lecture Notes in Computer Science*, pages 129–140, Santa Barbara, CA, USA, August 11–15, 1992. Springer Berlin Heidelberg, Germany.
- RB89. Tal Rabin and Michael Ben-Or. Verifiable secret sharing and multiparty protocols with honest majority (extended abstract). In *21st Annual ACM Symposium on Theory of Computing*, pages 73–85, Seattle, WA, USA, May 15–17, 1989. ACM Press.
- Sch99. Berry Schoenmakers. A simple publicly verifiable secret sharing scheme and its application to electronic. In Michael J. Wiener, editor, *Advances in Cryptology – CRYPTO’99*, volume 1666 of *Lecture Notes in Computer Science*, pages 148–164, Santa Barbara, CA, USA, August 15–19, 1999. Springer Berlin Heidelberg, Germany.
- SGA21. Jinhyun So, Basak Güler, and Amir Salman Avestimehr. Byzantine-resilient secure federated learning. *IEEE J. Sel. Areas Commun.*, 39(7):2168–2181, 2021.
- Sha79. Adi Shamir. How to share a secret. *Communications of the Association for Computing Machinery*, 22(11):612–613, November 1979.
- SS24. Victor Shoup and Nigel P. Smart. Lightweight asynchronous verifiable secret sharing with optimal resilience. *Journal of Cryptology*, 37(3):27, July 2024.
- SW19. Nigel P. Smart and Tim Wood. Error detection in monotone span programs with application to communication-efficient multi-party computation. In Mitsuru Matsui, editor, *Topics in Cryptology – CT-RSA 2019*, volume 11405 of *Lecture Notes in Computer Science*, pages 210–229, San Francisco, CA, USA, March 4–8, 2019. Springer, Cham, Switzerland.